

The joint entropy of two discrete random variables X and Y with joint distribution p is defined as

$$\mathbb{H}(X, Y) = -\mathbb{E}_{(X, Y) \sim p} [\log p(X, Y)] = -\sum_{x, y} p(x, y) \log p(x, y)$$

It quantifies the amount of uncertainty associated with the pair (X, Y) . The upper and lower bounds of the joint entropy are given by

$$0 \leq \max\{\mathbb{H}(X), \mathbb{H}(Y)\} \leq \mathbb{H}(X, Y) \leq \mathbb{H}(X) + \mathbb{H}(Y)$$

with later equality when X and Y are independent, in which $\mathbb{H}(X, Y) = \mathbb{H}(X) + \mathbb{H}(Y)$.

0.0.1 See also

0.0.2 References